



Atomic clock passes test

The Deep Space Atomic Clock could help spacecraft reach distant destinations on their own. <http://digit.in/jul21-49>



How Hans Berger invented EEG

How Hans Berger's quest for telepathy instead spurred modern brain science. <http://digit.in/jul21-50>

in remote areas “when you look at the transport of goods, let us say if you are transporting goods into the Himalayas in Arunachal Pradesh as an example, and the goods are being transported to a fairly remote part of the country. As they go through the Himalayas, quite often the signal is not available because that is a very difficult terrain. When let us say supplies are being transported, it needs to reach at a certain time, and there often a few things that can come into play.

The GPS location may or may not be available, but the communication is impossible without cellular network. There is no way the truck driver can communicate back and

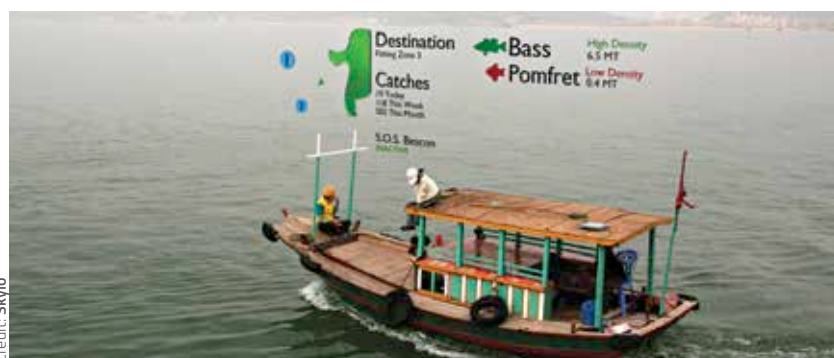
been tampered with. The truck can be geofenced so that it only moves along a specific route. All of these require data to be moving back and forth, but the quantum of data is not much. It is very very low throughput data that is required because its literally bytes and bits of data that needs to flow back and forth. The Skylo solution enables this and ensures 100 percent connectivity.”

All of this provides the fleet operators with the intelligence needed to make decisions, and it is at this stage where AI is being used to perform analysis on the data being captured. As the data is available as a continuous stream, the ability for organisations to respond to problems

lation, which is the only true broadband network available worldwide, is similarly designed and built to the very highest standards and is helping entire industries – those that have previously been unable to access reliable broadband services (such as maritime and aviation) – to transform themselves through the digital technologies our network enables.”

Inmarsat is gearing up to meet future capacity with a steady series of launches, with the schedule barely being disrupted by the pandemic, “Inmarsat has worked with its suppliers to ensure limited impact to spacecraft delivery dates due to the impact of Covid-19. We will be launching Inmarsat-6 F1 later this year on Mitsubishi Heavy Industries’ (MHI) H2A rocket. The Inmarsat-6s are the world’s most advanced commercial spacecraft ever built and Inmarsat’s first dual band (L and Ka-band) spacecraft. The Inmarsat-6s will ensure global L-band coverage until late 2030s. Inmarsat-6 F2 will be launched in 2022 and our five next generation, software defined Ka-band spacecraft in 2023 and 2024, including adding dedicated coverage for the Arctic region.

Inmarsat tends to launch big heavy satellites in higher geostationary orbit. A number of other internet connectivity companies are taking the approach of launching a large number of satellites, essentially spamming the low earth orbit with swarms of much smaller, cheap satellites. SpaceX, Telesat, OneWeb and Amazon are all taking this approach, with satellites in the Starlink constellation already appearing as trains of light in the sky. For example the Inmarsat-6 GEO satellites are each 7.5 meters high, and only 3 are needed to be deployed for global coverage. OneWeb’s LEO satellites are a meter high each, and hundreds are needed for global coverage. Mark explains how the Inmarsat is different, “Growth in the number of new space entrants has seen a significant increase in the number of



Skylo intelligence on a shipping vessel

forth in terms of what is happening, what are the road conditions, has there been a landslide on the way, is there some emergency services the truck driver needs and so on. For precious cargo, there is a lot of theft that happens on the way. Typically, it happens when there is no mobile network coverage because that is when the doors can be opened, the truck can be stopped for an hour and the truck driver can then say what could I do, its broken down and the truck cannot go forward.

There is no communication and the fleet operator, and all they can see if the GPS is functioning is that the truck is stuck at a certain location. Skylo’s solution provides realtime data to the fleet operator on the truck, the temperature of the cargo, state of electronic locks and if the locks have

increases, removing the need for guesswork or extrapolation.

Following the announcement, we had a chance to speak to Mark Dickinson, Vice President Space Segment of Inmarsat, about what makes their constellation rock solid. “Our L-band global network, which is being deployed by Skylo in India, is renowned as being the world’s most reliable satellite communications network. This is due to the exceptional engineering that goes into how we design and build our satellites, which in turn comes from the fact that millions of lives – at sea, in the air and on land - depend upon our satellites every hour of every day. Our network has been used to keep people safe for over 40 years and this translates into outstanding resilience. Our Ka band Global Xpress constel-